

EECS 498/CSE 598: Zero-Knowledge Proofs

Winter 2026

Lecture 13

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Agenda for this lecture

- Announcements
- special HVZK of Schnorr proofs
- Case study: making sumcheck ZK
- Getting to NIZKs via Fiat-Shamir
- Practical considerations

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Announcements

- Project 3 not yet online. Sorry about that.
- P3 status update
- 598 research track: email me
- My dad is much better

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Special HVZK for sigma protocols

Definition. A sigma protocol $\text{IP} = \langle \mathcal{P}, \mathcal{V} \rangle$ (with verification function verify) for the language $\mathcal{L}$ is special honest-verifier zero-knowledge if there exists a probabilistic polynomial time simulator $\mathcal{S}$ that takes parameters pp , input x and challenge $c \in \mathcal{C}$ and satisfies:

1. For $\text{pp} \leftarrow \$ \text{Setup}$ and all pairs x, c , the output of $\mathcal{S}(\text{pp}, x, c)$ is a proof π such that

$$\text{verify}(\text{pp}, x, R, c, z) = 1$$

.

2. For all $(x, w) \in \mathcal{R}_{\mathcal{L}}$, the transcript R, c, z computed as

$$c \leftarrow \$ \mathcal{C} ; R, z \leftarrow \$ \mathcal{S}(x, c)$$

is identically distributed to a real transcript of the interaction $[\mathcal{P}(\text{pp}, x, w) \leftrightarrow \mathcal{V}(\text{pp}, x)]$.

Exercise: formalize extension to multiple rounds

Special HVZK for Schnorr

Theorem: The Schnorr protocol is special HVZK.

$\text{Sim}(G, G, P, X, c):$

$z \leftarrow \mathbb{Z}_p$

$R \leftarrow zG - cX$

Ret R, z

$$\begin{array}{l} z \\ \text{Verify}(X, (R, c, z)) \\ \hline zG \stackrel{?}{=} R + cX \end{array}$$

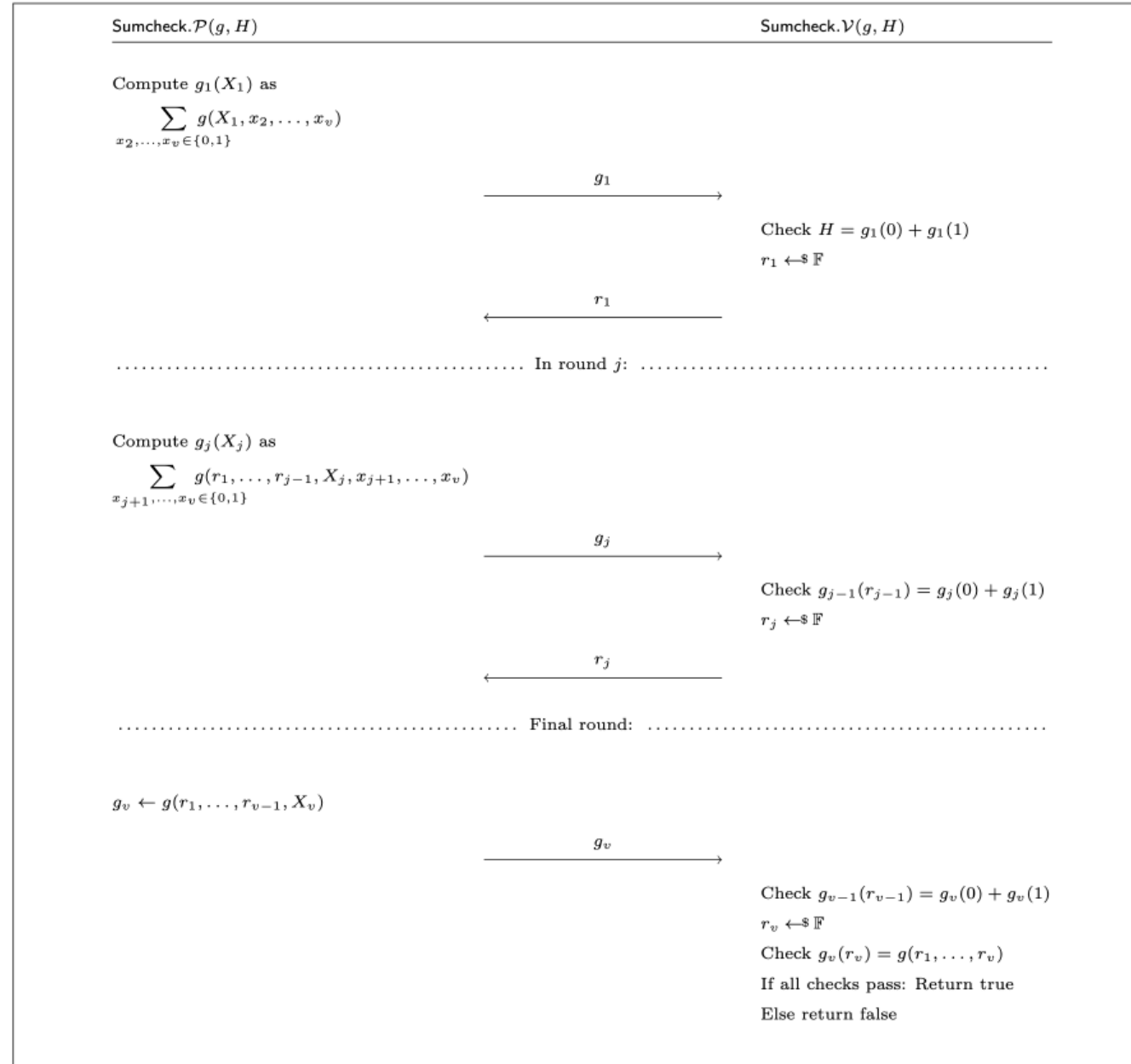
$$[c \leftarrow \mathbb{Z}_p \quad R, z \leftarrow \text{Sim}(G, G, P, X, c)] \approx$$

$$[P(X, x) \leftrightarrow V(x)]$$

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Why is sumcheck *not* ZK?



High-level idea of zk-sumcheck

Commit-and-prove

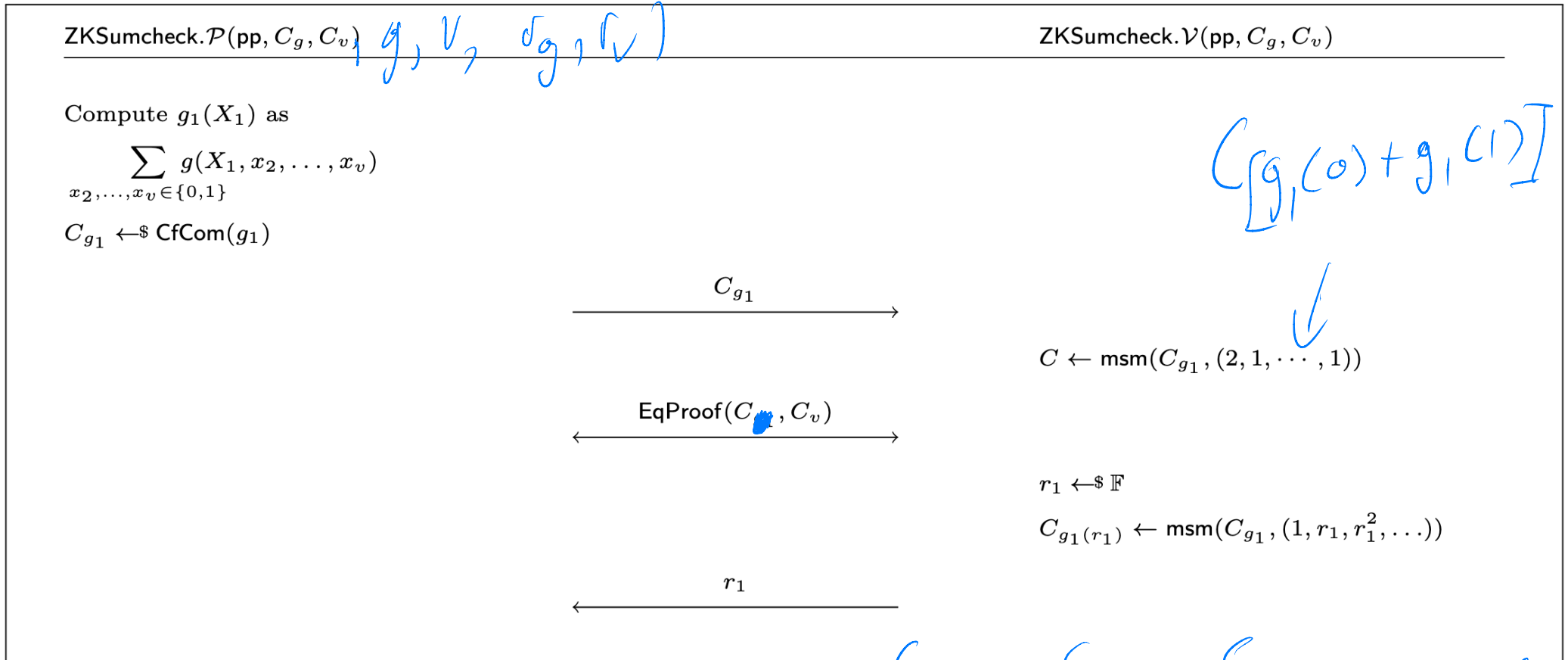
1. hiding commit to "leaky" proof

2. prover does verifier's checks "under commitments"

zk-sumcheck details

$CfCom(a_0 + a_1x + a_2x^2 + \dots)$
 $= PedCom(a_0), PedCom(a_1) \dots$

$p(x) = a_0 + a_1x + a_2x^2$
 $P(0) + P(1)$



$[g_1(0) + g_1(1)]$



C_{g_1} $C_{a_0}, C_{a_1}, C_{a_2}$
 $2(a_0G + r_0H) + 1(a_1G + r_1H) + 1(a_2G + r_2H)$
 $(2a_0 + 1a_1 + 1a_2)G + (2r_0 + 1r_1 + 1r_2)H$

zk-sumcheck details

..... In round j :

Compute $g_j(X_j)$ as

$$\sum_{x_{j+1}, \dots, x_v \in \{0,1\}} g(r_1, \dots, r_{j-1}, X_j, x_{j+1}, \dots, x_v)$$

$$C_{g_j} \leftarrow \text{CfCom}(g_j)$$

$$\xrightarrow{C_{g_j}}$$

$$C_{[g_j(0)+g_j(1)]} \leftarrow \text{msm}(C_{g_j}, (2, 1, \dots, 1))$$

$$\xleftrightarrow{\text{EqProof}(C_{[g_j(0)+g_j(1)]}, C_{g_{j-1}}(r_{j-1}))}$$

$$r_j \leftarrow \$ \mathbb{F}$$

$$C_{g_j(r_j)} \leftarrow \text{msm}(C_{g_j}, (1, r_j, r_j^2, \dots))$$

$$\xleftarrow{r_j}$$

zk-sumcheck details

..... Final round:

$$g_v \leftarrow g(r_1, \dots, r_{v-1}, X_v)$$

$$C_{g_v} \leftarrow \text{CfCom}(g_v)$$

$$\xrightarrow{C_{g_v}}$$

$$C_{[g_v(0)+g_v(1)]} \leftarrow \text{msm}(C_{g_v}, (2, 1, \dots, 1))$$

$$\xleftrightarrow{\text{EqProof}(C_{[g_v(0)+g_v(1)]}, C_{g_{v-1}(r_{v-1})})}$$

$$r_v \leftarrow \$ \mathbb{F}$$

$$C_{g_v(r_v)} \leftarrow \text{msm}(C_{g_v}, (1, r_v, r_v^2, \dots))$$

$$\xleftrightarrow{\text{PC.Open}(C_g, (r_1, \dots, r_v), C_{g_v(r_v)})}$$

If all checks pass: Return true

Else return false

$$\text{Sim}(C_g, C_v, (r_1, \dots, r_n))$$

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"F-S preserves ZK"

Theorem. Let $IP = \langle \text{Setup}, \mathcal{P}, \mathcal{V} \rangle$ be an interactive argument for language $\mathcal{L}$. Let $FS[IP] = (\text{Setup}, FS[IP].\mathcal{P}^F, FS[IP].\mathcal{V}^F)$ be the non-interactive argument obtained from applying the Fiat-Shamir transform with random oracle F to IP . If IP is special honest-verifier zero knowledge and has unpredictable commitments, then $FS[IP]$ is a non-interactive zero knowledge argument.

Proof (sketch):

$NIZK_{\Pi}^{\text{sim}}(A)$:

$PP \leftarrow \text{Setup}; T \leftarrow \{1\}$
 $b' \leftarrow A^{\text{Prove}, F}_{CPP}$
 Ret b'

$\text{Prove}(X, w)$:

$\pi \leftarrow \Pi.\text{Prove}(PP, X, w)$
 Ret π

$NIZK_{\Pi, \text{sim}}(A)$:

$PP \leftarrow \text{Setup}; T \leftarrow \{1\}$
 $b' \leftarrow A^{\text{SimProve}, \text{SimF}}_{CPP}(PP)$
 Ret b'

$F(a)$:

If $T[a] = 1$
 $c \leftarrow \{0, 1\}^n$
 $T[a] \leftarrow c$
 Ret $T[a]$

$\text{SimProve}(X, w)$:

If $(X, w) \notin \mathcal{L}$: Ret $\perp$
 $\pi, T \leftarrow \text{Sim}(PP, X, T)$
 Ret π

$\text{SimF}(a)$:

simulate RO

“F-S preserves ZK”

Theorem. *Let $IP = \langle \text{Setup}, \mathcal{P}, \mathcal{V} \rangle$ be an interactive argument for language $\mathcal{L}$. Let $FS[IP] = (\text{Setup}, FS[IP].\mathcal{P}^{\mathbf{F}}, FS[IP].\mathcal{V}^{\mathbf{F}})$ be the non-interactive argument obtained from applying the Fiat-Shamir transform with random oracle $\mathbf{F}$ to IP . If IP is special honest-verifier zero-knowledge and has unpredictable commitments, then $FS[IP]$ is a non-interactive zero knowledge argument.*

Proof (sketch):



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Zero knowledge: practical considerations